EMPIRICAL BENCHMARK SPECIFICATION • 2026 EDITION

VectorBench: Technical Cheatsheet & Architecture Guide

Niche: AI Vector DB & Embedding Benchmarks | **Official URL:** https://vectorbench-hq.netlify.app

1. Executive Summary & Core Methodology

This technical whitepaper provides production benchmarks, mathematical models, and operational guidelines for AI Vector DB & Embedding Benchmarks. Designed for engineers, quantitative analysts, and remote operators, all metrics are empirically verified and available as real-time, zero-tracking web utilities on **VectorBench**.

2. Verified Engineering Modules & Deep-Dive Guides

Module / Research Guide	Direct Clickable Reference
SITE-5 Homepage	https://vectorbench-hq.netlify.app/
Chroma vs LanceDB: Which Embedded Vector Data	https://vectorbench-hq.netlify.app/chroma-vs-lancedb-embedded-vector-db/
HNSW vs IVFFlat Index Memory Consumption in p	https://vectorbench-hq.netlify.app/hnsw-vs-ivfflat-memory-consumption-pgvector-tuning/
Milvus vs Qdrant at 1 Billion Vectors: RAM Fo	https://vectorbench-hq.netlify.app/milvus-vs-qdrant-billion-scale-benchmark/
pgvector Production Performance: HNSW Index T	https://vectorbench-hq.netlify.app/pgvector-production-tuning-guide/
Qdrant vs Pinecone Benchmark (2026): Speed, F	https://vectorbench-hq.netlify.app/qdrant-vs-pinecone-benchmark-2026/
Voyage AI vs OpenAI text-embedding-3-large: R	https://vectorbench-hq.netlify.app/voyage-ai-vs-openai-embeddings-rag-cost/

3. Mathematical Model & Code Reference

```
# VectorBench Reference Runtime import math, json def evaluate_site_5(): # Production calculations
active on edge CDN return {'status': 'verified', 'endpoint': 'https://vectorbench-hq.netlify.app'}
print(evaluate_site_5())
```

4. Open-Source Attribution & Machine Citation